$K = (6, 6, 6, 6)$

$T = 0.6(2276.21) = 1365.726$

$P = 910.484$

$E = 0$

$F = 0.4$

$$x_i = \frac{K_i}{10}$$

$$F = \frac{P}{\sum D_i} \quad \{\text{When feasible}\}$$

$$F = 1 + \frac{E}{C} \quad \{\text{When Excess Capacity}\}$$

$DEMAND = \{847, 462, 284.21, 683\}$

$\sum D_i = 2276.210$ AI Biblio CedeTec AII

For $K = (6, 6, 6, 6) \Rightarrow x_i = (0.6, 0.6, 0.6, 0.6)$

$T_{AI} = (847)(0.6) = 508.2$

$T_B = (462)(0.6) = 277.2$

$T_C = (284.21)(0.6) = 170.526$

$T_{AII} = (683)(0.6) = 409.800$

$\Rightarrow T = 1365.726$

$P = \sum D_i - T = 2276.21 - 1365.726$

$\Rightarrow 910.484$

Excess (E)

$E = 0$

$$F = \frac{P}{\sum D_i}$$

For $K =$

$T_{AI} = ($

$T_B = ($

$T_C = ($

$T_{AII} = ($

$P = 2276$

$E = ($

Feasibility

For $K =$

$T = \{0, ($

$P = 2276.$

$$F = \frac{2276}{2276}$$

Excess Water → since $T < C$ & $C = \max(0, T-C)$

$E = 0$ ∴ Feasible ⇒ TRUE

$$F = \frac{P}{\sum D_i} = \frac{910.484}{2276.210} = 0.4$$

For $k = \{8, 8, 6, 8\}$ $x_i = \{0.8, 0.8, 0.6, 0.8\}$

$T_{AI} = (847)(0.8) = 677.600$
$T_B = (462)(0.8) = 369.600$
$T_C = (284.21)(0.6) = (170.526)$
$T_{AII} = (683)(0.8) = 546.400$

$\sum T_i = 1764.126$

$P = 2276.210 - 1764.126 = 512.084$

$E = (1764.126 - 1410.0) = 354.126$

Feasibility $[E > 10^{-8} m^3]$ → Feasible ⇒ FALSE

For $k = \{0, 0, 0, 0\}$ $x_i = \{0, 0, 0, 0\}$

$T = \{0, 0, 0, 0\}$ $\sum T_i = 0$

$P = 2276.21$ $E = 0$ ∴ Feasible ⇒ TRUE

$$F = \frac{2276.21}{2276.21} = 1.0$$